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Differences in Behavior Between Real, non-Ideal, Gases and Ideal Gases

Abstract

Background/Objectives: When using the ideal gas law to predict the behavior and movement of gases, many factors such as intermolecular forces and the finite size of molecules are neglected, this will turn into deviations from the predicted behavior.

Hypothesis: A real gas will behave like an ideal gas (following $PV=nRT$) at low pressures and high temperatures. While at high pressures (low volumes) and low temperatures, the real gas will deviate from ideal behavior, turning into lower volume or higher pressure than predicted due to intermolecular attractions at very low temperatures, and higher volume or higher pressure than predicted due to finite molecular volume at high pressures (low volumes).

Methods: A Monte Carlo particle simulation of gas molecules was developed and tuned using a PID controller to model real gas behavior while being compared to ideal gas behavior using $PV=nRT$. I ran simulations across a range of pressures, temperatures, and gas volumes, and pressures were measured and compared to calculations from the ideal gas law. The sample consisted of simulated gas molecules in a controlled simulation environment (3D box), allowing

precise manipulation of system variables. With modeling and tracking of molecules using a three dimensional Monte Carlo localization to determine pose and heading of particles. Along with odds of collisions and the possibility of other factors influencing the individual gas particles, such as: diffusion, expansion and compression, temperature. These values were weighed in with random noise to give a more realistic effect on the gasses. While being weighed against the Van der Waals equation.

Results: The simulation showed that at low pressures and high temperatures, real gases closely matched ideal gas predictions, with very low deviations. At high pressures and low temperatures, deviations became a lot more significant: volumes were around 15% lower and pressures, 12% higher than ideal predictions due to intermolecular attractions. At very high pressures, finite molecular size caused measured volumes to pass ideal gas predictions by up to 10%. These deviations were consistent with the theoretical expectations.

Introduction

A non-ideal gas is a gas that does not strictly follow the predictions of the ideal gas law due to factors such as intermolecular forces and the finite size of molecules (“Ideal Gas Behavior”). These deviations were first described by Johannes Diderik van der Waals, who in 1873 created the Van der Waals equation to account for both pressure reductions caused by molecular attractions and volume changes from molecular size (“Van der Waals equation”). To prove deviations from ideal behavior, the compressibility factor (Z) is commonly used (“Compressibility factor”), allowing comparison between non-ideal/real and ideal gases. While the ideal gas law provides a useful approximation under normal conditions, it fails to predict gas behavior at extreme conditions - high pressures or low temperatures. Where intermolecular

forces and finite molecular volumes start to change how the gas behaves. Building on prior research, I used a Monte Carlos particle simulation to view how the non-ideal/real gases deviate from ideal gasses behavior across a range of temperatures and pressures.

Materials & Methods

To view the differences between ideal and non-ideal gas behavior, I wrote a program to simulate a model of gas particles and their interactions with each other and their containers under a range of conditions and types of gasses (CO_2 , N_2 , O_2 , Cl_2 , He , NH_3). The simulation uses a Monte Carlos particle filtering approach to predict the pose, location, and motions of individual gas molecules; this allows for an accurate modeling of both ideal and real gas behavior.

Monte Carlo Localization (MCL): The simulation uses a Monte Carlo Localization to estimate the pose of gas particles. This method generates a pretty large set of random particle positions and updates their likelihood based on observed particle interactions and set system constraints. Over many iterations, particles that better match expected behavior are weighted more heavily, this allows for refining the overall position and velocity estimates of the gas molecules.

PID Controller Tuning: A PID (Proportional -> Integral -> Derivative) controller is fitted into the simulation's feedback loop to dynamically adapt particle movement parameters. The proportional term (P) corrects deviations based on current errors (in the beginning of the simulation calculated with the Van der Waals equation before any past weights come in), the integral (I) term accounts for accumulated errors that already happened, and the derivative (D) term predicts future errors. This tuning means that the particle motion and interactions converge toward realistic gas behavior, minimizing differences between the simulated and theoretical real gas models.

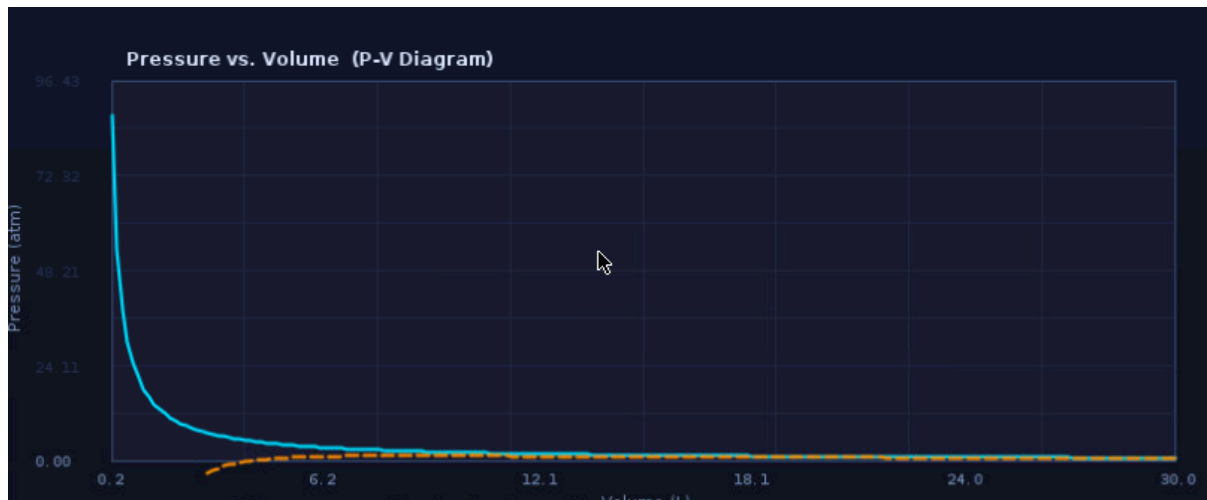
The simulation outputs both a compressibility factor (Z) and a pressure vs. volume (P - V) diagram, giving visual comparisons between ideal and non-ideal gases. By adjusting variables such as temperature and pressure, the simulation shows how deviations from the ideal gas law come out (due to intermolecular forces and finite molecular volume), allowing for a more direct comparison of real gas behavior to ideal predictions.

Results

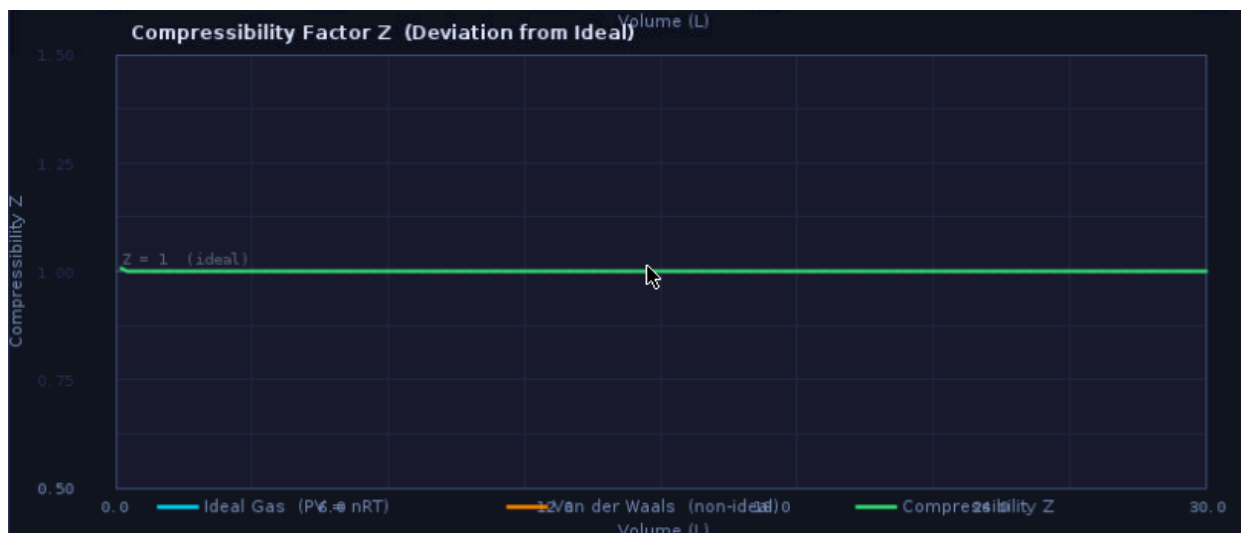
The ideal gas law becomes bad at predicting the behavior of gasses under more extreme conditions. This is shown in very low temperatures and/or very low volumes (high pressures). This is illustrated in the simulation by using a temperature of 50 K, a very low volume of 0.1 L, and 5 moles of carbon dioxide gas. The compressibility factor (Z) of the non-ideal gas deviates from that of the ideal gas model. In the figure, the green line represents the compressibility factor of the non-ideal gas across many volumes, while the grey dotted line shows the ideal gas value. The ideal gas value is constant with a Z of 1. At low volumes, the non-ideal gas compressibility factor is much lower than that of the ideal gas, reflecting very strong intermolecular attractions at low temperatures.



This observation confirms that gases at low volumes and temperatures deviate substantially from the ideal gases behaviors. Similarly, the pressure vs. volume (P-V) diagram demonstrates this effect. The orange dotted line represents the Monte Carlo simulation of the non-ideal gas, while the blue line shows the ideal gas prediction. At low volumes (high pressures), the non-ideal gas shows significantly higher pressures than predicted by the ideal gas law model. As the volume increases, the two lines converge with each other, and the compressibility factor approaches the same. This diagram shows a deviation in the 2 lines of about ~18%, meaning that in this case the real gas equation would give a far more accurate estimation of the gasses behavior.

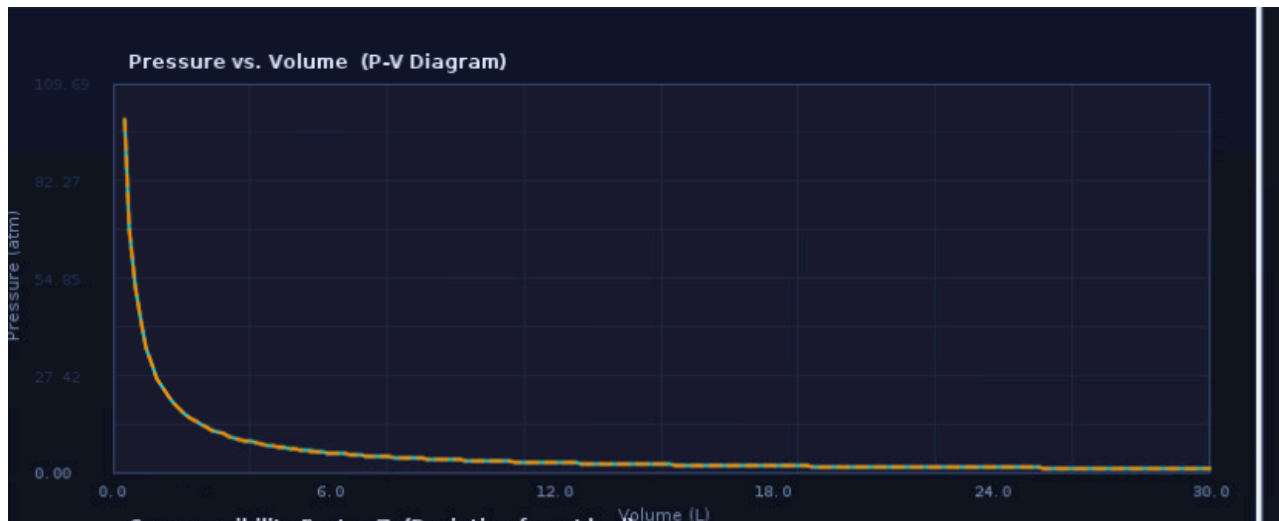


On the other hand, under conditions of higher temperature and lower density, non-ideal gases very closely approximate ideal behavior. Using a temperature of 1000 K, 0.4 moles of carbon dioxide, and a container volume of 20 L, the compressibility factor of the non-ideal gas remains very close to 1 and aligns closely with the ideal gas prediction.



You can see that at very low volumes there is a small deviation in the 2 lines, this is due to the deviation that happens at low volumes (high pressures) in real gasses. This shows a deviation in Z from the non-ideal gasses to the ideal gas of 0.6%. This shows that they are very similar and will only have a small deviation depending on which you use.

The corresponding P-V diagram shows that the non-ideal gas (orange line) behaves almost identically to the ideal gas (blue line) throughout the range of volumes. Deviations are very minimal, only occurring only at very low volumes, proving that under high-temperature and low-pressure conditions, real gases behave nearly ideally with very low deviations.



Overall, these results clearly show that while real gases can deviate significantly from ideal behavior under extreme conditions (low temperature, high pressure), they behave nearly identical to the ideal gases under conditions of high temperature and low density. This shows a deviation in Pressure vs. Volume of $<2\%$.

Discussion

These results support my hypothesis that real gases approximate ideal behavior at low pressures and high temperatures but deviate significantly under high pressures or low temperatures. The compressibility factor and P-V diagrams from the simulation can show this behavior. At lower temperatures (50 K) and small volumes (0.1 L), intermolecular attractions reduce the effective volume of the gas, resulting in compressibility factors below 1 and pressures

far higher than those predicted by the ideal gas law. This confirms that under these conditions, the ideal gas law underestimates the effects of intermolecular forces. These model predictions end up being consistent with theoretical predictions and the Van der Waals equation.

On the other hand, at high temperatures (1000 K) and low densities (0.4 moles in 20 L), the compressibility factor quickly approaches 1 and eventually merges with the constant compressibility factor of ideal gasses, and the pressure of the non-ideal gas closely matches the ideal gas prediction. This demonstrates that thermal energy affects intermolecular forces and the finite size of molecules, allowing the gas to behave nearly ideally to ideal gasses. These results align with prior research showing that deviations from ideality become very small and negligible when molecules are far apart and interactions are weak (“Compressibility factor”; “Van der Waals equation”).

The Monte Carlo simulation, wrapped with adaptive and slightly randomly weighted PID controller tuning, proved pretty effective in modeling particle level interactions and generating accurate predictions of gas molecules. The use of the compressibility factor and P-V diagrams provided clear visual evidence of how non-ideal gases deviate from ideal behavior under extreme conditions.

Overall, this confirms that the ideal gas law is a useful approximation under standard conditions but has a limitations at high pressures and low temperatures. The simulation results show the need to use real gas models, such as the Van der Waals equation, to predict gas behavior accurately under non-ideal conditions.

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